

Service Manual

DOC223.52.03913.Apr04 - Changes reserved

>>SC PROBES

Solitax





This service manual is intended to provide assistance during all regular service work as well as other repairs. Any related software menu is shown in full in the related operating instructions. Service related extracts of this menu are repeated here.

The latest version of this service manual can and should be downloaded from the intranet.

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1. Inspection and repair instructions

1.1 Checking tank edge mounting

The correct mounting of the probe is to be checked.

- Alignment parallel to the flow
- Immersion depth
- Distance to the tank base and tank wall
- If necessary protection against pebbles, branches, etc.

1.2 Service menu

→ SENSOR SETUP

- + **Calibrate** (*instrument factor, offset, reset to factory calibration, etc*)
- + **Configuration** (*location, parameters, logger interval etc*)
- + **Test/Maint**
 - **Probe Info**
 - **WIPER PROFILE**
 - **Counter**
 - **Maint. Proc.**

Along with the functions that the customer can open using the controller (wiper counter reset) there are further functions that can only be opened by technical customer service. For this purpose it is necessary to enter the password

MAX100 (*currently only on sc100*)

1.3 Software Updates

The related software can always be downloaded from the HACH-LANGE intranet (SERVICE/SOFTWARE/...).

The following is available:

- **sc instrument software** (current software for sc components)
- **sc sensor flash** (program for updating the sc100 display unit)
(instructions in the form of a pdf file are also included here)

1.4 Calibration

(Special) tools necessary: sc100 or sc1000 controller
formazine standard,

Preparation

a) Pour formazine standard into a suitable container and dilute such that a value of >600 FNU is produced. Measure this value with another instrument (Nephla, Hach, other Solitax probe).

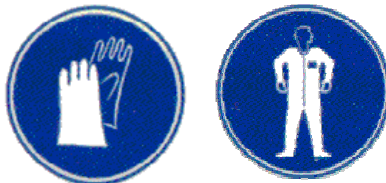
OR

b) Dilute defined formazine standard, e.g. 4000 FNU standard to 500 FNU (note life of formazine standard!)

When handling formazine:
avoid contact with the skin!

If necessary, **settings on the Nephla:**

- Integration 0-3
- Dimension FNU
- Correction 1.0
- RS232 PC



The calibration of the probes is supported by the sc100 and sc1000 controllers with prompts:

- MENU
- SENSOR SETUP
- TEST/MAINT
- MAINT. PROC.
- ENTER
- Password entry: MAX100

Activity/ menu prompt	Comment
Window clean?	
Immerse probe in water (dist.) Determine zero point values N1 and N2 [Nom.data N1: 4-10] [Nom.data N2: 31-47]	Use black container! Rotate probe slowly so that any air bubbles can escape! (zero point values: LED switched off)
If values OK (stable): ENTER	Software only accepts values in the nominal range.
[Nom.data LED: 100-800] [Nom.data N1OP: 0-20]	Optical zero point and "LED" are determined (LED switched on)
If values OK (stable): ENTER	Software only accepts values in the nominal range.
Place probe in container with defined turbidity	
Measure turbidity of the formazine solution/enter using keypad, ENTER	See above
Indication of F1 and F2, accept with ENTER [Nom.data F1: 0,3 - 5] [Nom.data F2: 0,3 - 5]	- Rotate probe slowly so that any air bubbles can escape! - During the measurement, maintain minimum distance to wall of the container! - Variation in the last digit of the measured value is allowed - Software only accepts factors in the nominal range.
Clean probe	
Finished!	
The calibration should then be checked by measuring a turbidity standard with a significantly different turbidity (and (dist.) water).	Read turbidity on the display unit

1.5 Probe housing and internal layout

- Seal replacement

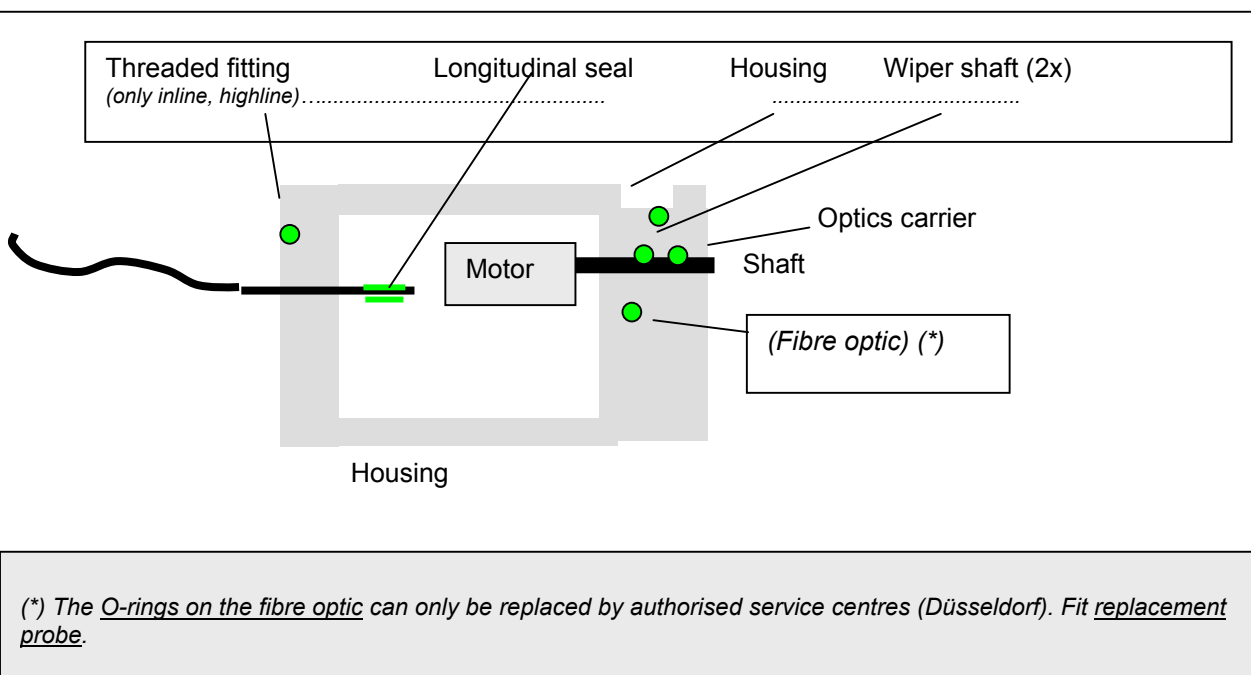


Important note:

It must be ensured that the optical components, some of which are highly sensitive, are not damaged by soiling.

The active optical surfaces on all optical parts used are to be kept **clear of soiling** (gloves!).

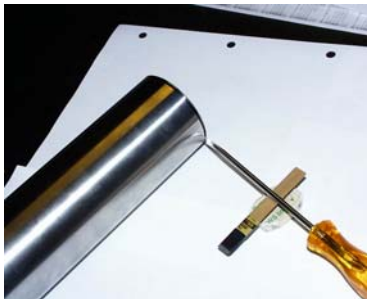
Always only **place** the **optics carrier on soft material** during servicing work, **as otherwise the window may be seriously damaged.**



1. Opening the housing

- For all probes: undo the cable gland
- Open the housing: varies depending on the type

inline and highline



Bolts screwed in 4-5 turns cannot be completely unscrewed, open housing by applying pressure to the bolt heads

hs-, ts-line and t-line

Unscrew 3 setscrews (1mm AF)

- Carefully remove housing



Attention:

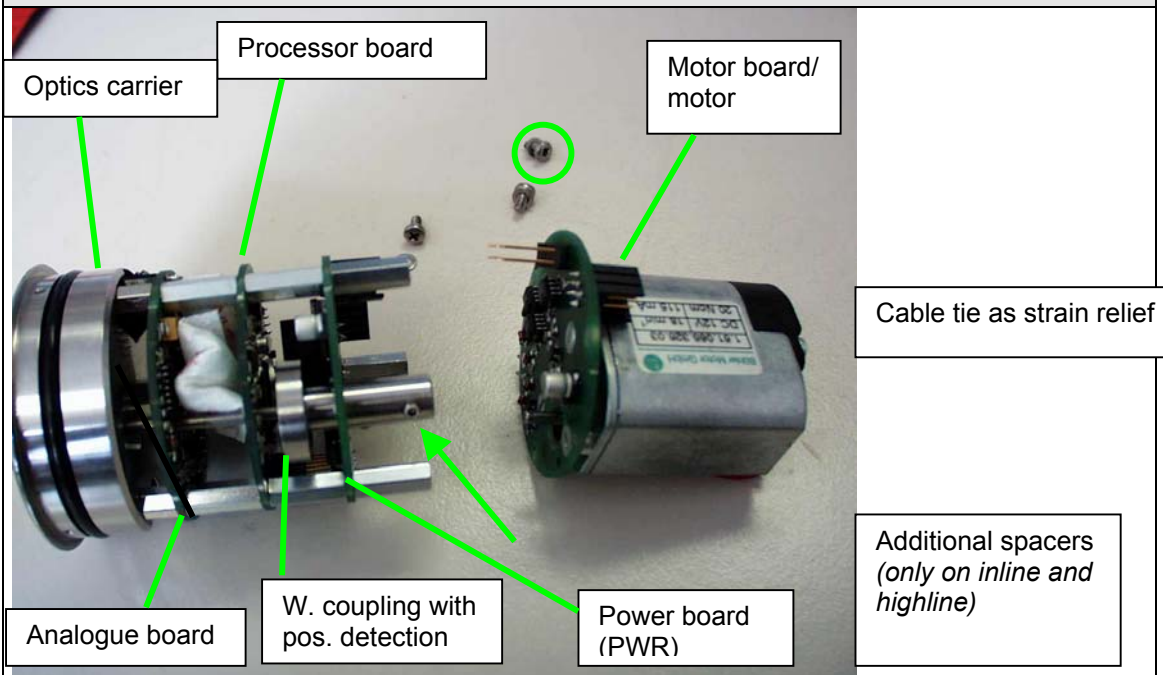
For subsequent dismantling stages it is recommended to re-connect the probe to a controller and to activate the **wiper** using MAINT. PROC. as necessary.

Lock the wiper manually when a position is reached in which the two **setscrews for the coupling** can be accessed.

Then disconnect the connector again!

2. Dismantling the probe

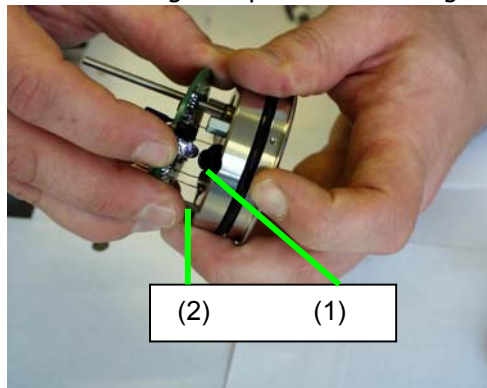
After removing 3 bolts and undoing the wiper shaft coupling, it is possible (if necessary) to pull off the **motor board with motor**.



The following can be dismantled one after the other:

- Long spacers (on inline and highline) with cable guide and 3 cylinder head bolts on ts-line and t-line
- **Power board**
- **Shaft coupling with position detection**
- **Processor board**

After removing the spacers the analogue board can be dismantled:

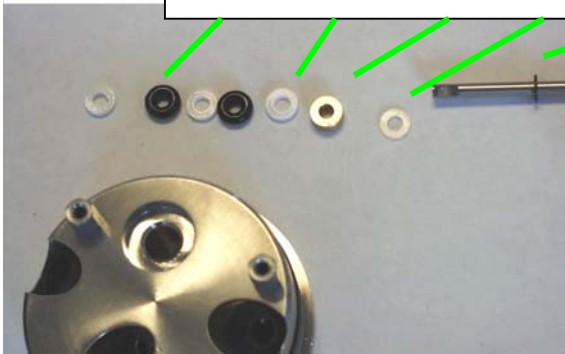


Using a finger, carefully pull the light emitting diode (1) out of its bore, the photodiodes (2) are pulled out at the same time.

Dismantle wiper blade

Dismantling the **wiper shaft** (replacement)

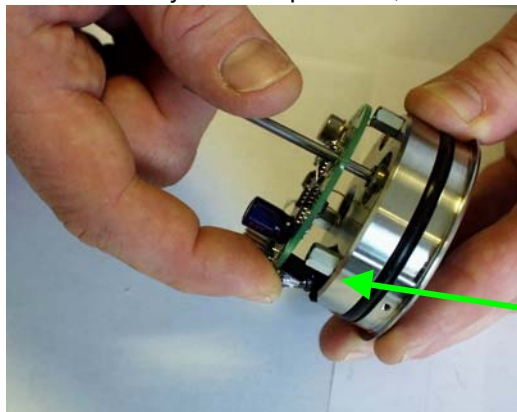
Installation:
2 O-rings – 3 bearing rings – Sleeve - Washer - Locking washer



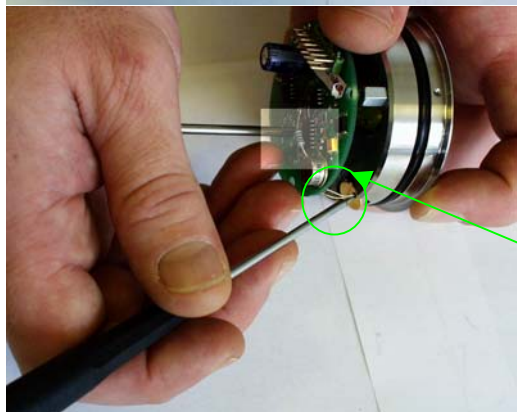
Dismantle old O-rings and bearing rings; clean grooves and shaft
Grease new O-rings sparingly and fit as above, fit bearing rings as above
Dismantle housing seal, clean grooves
Grease new O-rings sparingly and fit
Re-fit shaft

Assembly

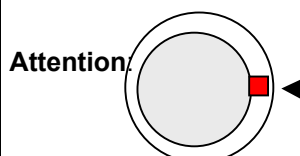
After assembly of the wiper shaft, the analogue board can be fitted:



With a finger carefully press the light emitting diode into its bore, until it engages with a "click". Alignment with photocell OK?



Carefully press the two photodiodes into their bores until you feel them noticeable engage.



The lugs on the photodiodes must **not be in contact** with the housing after assembly!

Fit processor board

- Fit PWR board
- Align coupling with pos. detection on shaft and fix
- Fit **wiper blade** with **new setscrew**
- If necessary fit motor board with motor, fix coupling

If necessary replace the motor



When re-fitting the motor, pay attention to the 3 spacers and the securing of the cable to the controller display unit with cable ties. Cable guide!

Insert **desiccant**, check **longitudinal seal** for the connecting cable!

Replace O-rings under threaded fitting (only on *inline* and *highline*)

Bolt together housing, tighten cable gland

Calibrate probe

1.6 Cleaning the measuring window

Always place the optics carrier on soft material during servicing work, as otherwise the window may be seriously damaged.

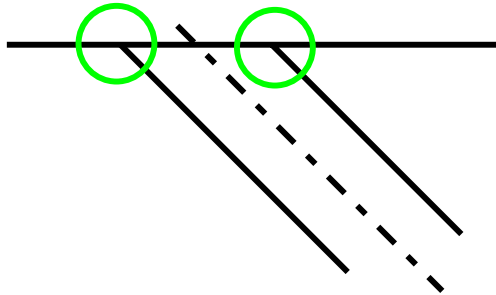
As a rule the window can be cleaned with conventional cleaning agent and a cloth.
In case of very stubborn deposits, 10% hydrochloric acid can be used.

Attention:



Checking:

The fibre optic must be exactly flush with the surface of the optics carrier!



1.7 Inspection and repair log

	Work within the Service contract	
	Every 12 months (or on reaching the maximum)	Every 24 months
PROBE		
♦ Check mounting	O	
♦ <i>Operator: replace wiper profile, when > 10,000 (max. 20,000)</i>	O	
♦ Read number of drive motor cycles, replace when nominal value (1 million) exceeded	O	
♦ Replace housing seals, if necessary replace shaft seals		O
♦ Clean measuring window	O	
♦ Calibration	O	
♦ Renew desiccant		O
Controller		
Reset counter for the maintenance work performed	O	
PIPE INSTALLATION FITTING (if used)		
♦ Grease chain drive	As required	

2. Troubleshooting

2.1 Messages

WARNING MESSAGES		ERROR MESSAGES
The probe remains in the operating state.		The probe is placed in the service state.
Replace Wiper		Pos. unknown
Test/Maint		LED Failure
Gasket		Moist
		Cal. Data

(The possible messages are identical for all probes.)

2.2 Actions

Warning messages

Error message	Causes		Actions
Replace Wiper	A	Counter elapsed	Perform

Error message	Causes		Actions
Test/Maint	A	Counter elapsed	Perform

Error message	Causes		Actions
Gasket	A	Counter elapsed	Replace seals

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Error messages

Error message	Causes		Actions
Pos. unknown	A	Wiper position unknown	Check wiper control Check position detection

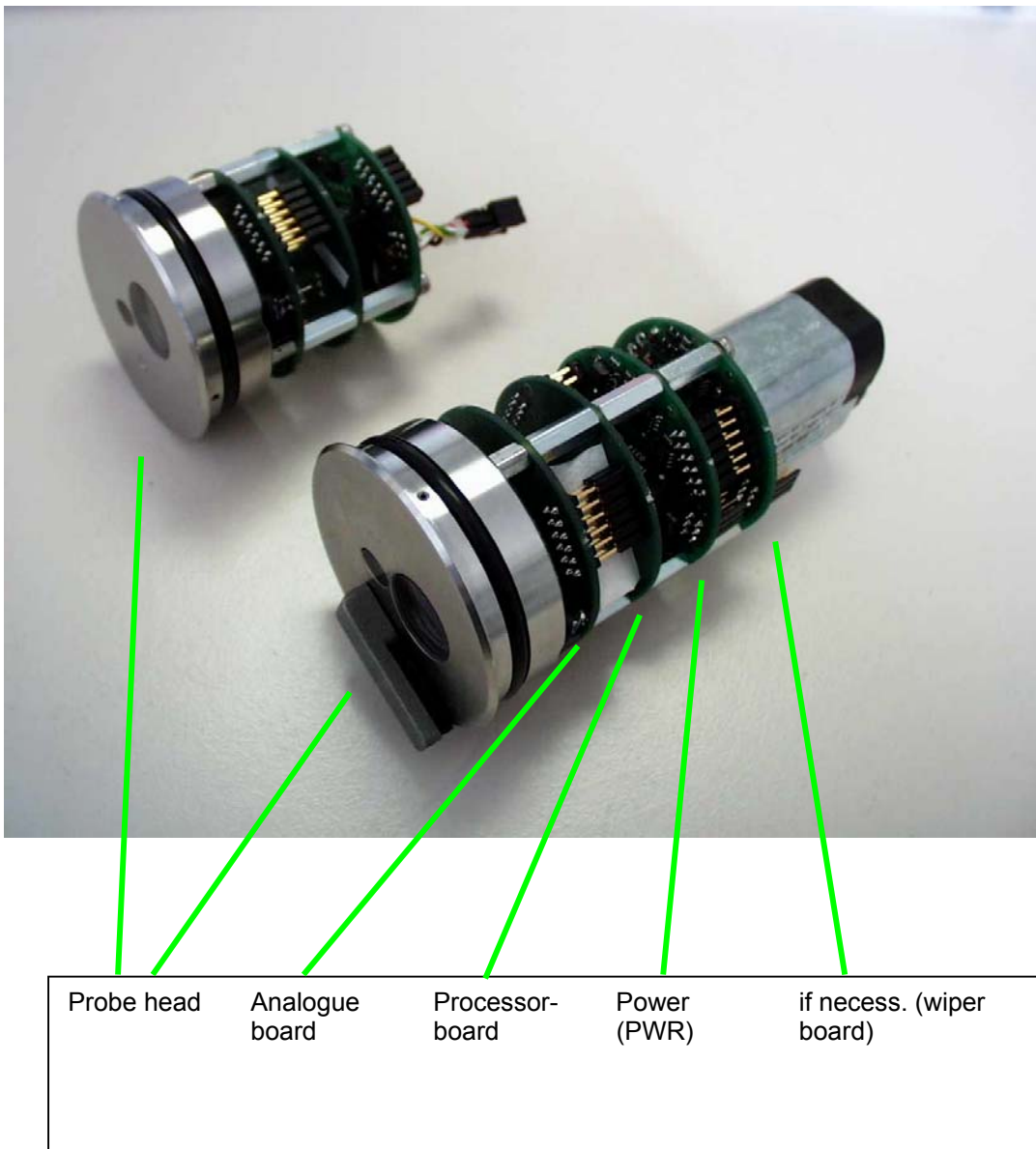
Error message	Causes		Actions
LED Failure	A	Faulty LED: less than 10% of the original intensity is measured by the REF channel	Replace analogue board

Error message	Causes		Actions
Moist	A	Moisture value > 10	Check seals, optics carrier, cable gland and housing

Error message	Causes		Actions
Cal. Data	A	Factory calibration data lost (after the replacement of the processor board)	Perform calibration

3. Assemblies

3.1 Housing layout



4. Variants

4.1 Hardware versions

	Stainless steel housing	Plastic housing
Immersed probes		
t-line (Only turbidity from 0.001 to 4000 FNU)	---	With wiper: LXG423.99. 10000 Without wiper: LXG423.99. 12000
ts-line (Turbidity and TS from 0.001 to 50 g/l)	With wiper: LXG423.99. 00100 Without wiper: LXG423.99. 02100	With wiper: LXG423.99. 10100 Without wiper: LXG423.99. 12100
hs-line (Turbidity and TS from 0.001 to 150 g/l)	With wiper: LXG423.99. 00200 Without wiper: LXG423.99. 02200	With wiper: LXG423.99. 10200 Without wiper: LXG423.99. 12200
Probes installed in pipes		
Inline (Turbidity and TS from 0.001 to 50 g/l)	With wiper: LXG424.99. 00100 Without wiper: LXG424.99. 02100	---
Highline (Turbidity and TS from 0.001 to 150 g/l)	With wiper: LXG424.99. 00200 Without wiper: LXG424.99. 02200	---

4.2 Software versions

Version	Date	Comment/changes
1.0	April 2004	Launch

5. Consumables and spare parts

Spare parts list
(date 8.4.04)

EVQ125	Hand crank	ELESA GN 570.1-64-V8
HYT046	Case for SOLITAX probes	
LZX050	Wiper profiles for SOLITAX 5 off	Made of silicone
LZX455	Optics carrier, stainless steel	
	Optics carrier, plastic	
LZX059	Wiper shaft	
	Probe cable, comp.	
	Probe housing, stainless steel	
	LXG423	
	Probe housing, stainless steel	
	LXG424	
	Probe housing, plastic LXG423	
LZX077	Geared motor	
	Solitax SC program YYX518	Probe program
	Solitax SC program YYX519	Database
LZX223	Wiper comp.	
LZX302	Desiccant (60 x 105 mm)	
LZX337	Pipe installation safety fitting for	SOLITAX probes installed in pipes
LZX414.00.10000	Probe bracket 90° adapter	
LZX417	Set of hardware for probe	
LZX421	Set of seals for Solitax probes	
LZX771	Set of bolts SOLITAX SC	
LZX423	Set, optics carrier for Solitax	(filter, shaft...)
Cf. sc1000	5m extension cable for	probes, sensors
Cf. sc1000	10m extension cable for	probes, sensors
Cf. sc1000	15m extension cable for	probes, sensors
Cf. sc1000	20m extension cable for	probes, sensors
Cf. sc1000	30m extension cable for	probes, sensors
Cf. sc1000	50m extension cable for	probes, sensors
Cf. sc1000	50m extension cable for	probes, sensors
Cf. sc1000	100m extension cable for	probes, sensors
LZX461	Pipe installation fitting for	SOLITAX probes installed in pipes
LZX578	Wiper profiles for SOLITAX 5 off	Made of Viton
XLT560	Cable gland PG 9	15 long for cable thickness 6.5 - 4.0
	Analogue board	ZBA767
	Processor board	ZBA765
	Power board	ZBA766
	Wiper board	ZBA800

6. What's New

Change/addition since last issue	Page	Comment
Nominal data given	Chap 1.4	